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Code Explanation

The program evaluates 100000 terms of an alternating series, meaning the terms switch between positive and negative signs. Each denominator is an odd number of the form 1, 3, 5, 7, and so on, so the program builds the sequence of reciprocals of odd integers with alternating signs. The summation is carried out term by term and then multiplied by 4, because the infinite Gregory-Leibniz relation states that the inner sum equals $\pi/4$. This structure makes the program mathematically simple.

Result of the Analysis

The result of the experiment is a numerical value close to 3.14159, which shows that the finite partial sum gives a reasonable approximation of π .

Discussion

This experiment shows that an alternating series can be used to approximate an important mathematical constant. The approximation improves as more terms are added because the term sizes decrease, which is a key condition behind convergence for alternating series.